**Ben-Gurion University of the Negev**
Electrical & Computer Engineering

**Advanced CPU Architecture & Hardware Accelerators**

Course 361-1-4693

# LAB 5 — Preparation Report

## Scalar Single-Cycle & 5-Stage Pipelined RV32IM CPU µArchitecture

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# 1. Aim and Assignment Definition

This report documents the design, synthesis, implementation and validation of a RISC-V compatible CPU on an Intel FPGA, in two micro-architectural variants that share the same RV32IM (MULDIV-partial) instruction set:

- Part 1 — a scalar single-cycle RV32IM core, obtained by elevating the supplied single-cycle RV32I datapath with the M-extension mul instruction.
- Part 2 — a scalar 5-stage pipelined RV32IM core, obtained by slicing the single-cycle datapath into IF / ID / EX / MEM / WB stages with full data-forwarding, combinational hazard (interlock) detection, and branch/jump resolution in stage 4.

Both cores follow a Harvard organisation with separate ITCM (instruction) and DTCM (data) tightly-coupled memories of 8 kB each, a standard RISC-V register file, and a structural top level. A single PLL-derived clock (MCLK) drives the design; push-button KEY0 is the active core reset that brings the PC to the first program instruction.

## 1.1 The mul instruction and the 16-bit multiplier

From the M-extension only mul rd, rs1, rs2 is required. It multiplies the lower 16-bit half-words of rs1 and rs2 and writes the 32-bit product to rd. The 16×16 multiplication is built from the FPGA's embedded 8-bit multipliers using four partial products:

- $P0 = A\_low \times B\_low$, $P1 = A\_low \times B\_high$, $P2 = A\_high \times B\_low$, $P3 = A\_high \times B\_high$
- $M = P1 + P2$, $RESULT = P0 + (M \ll 8) + (P3 \ll 16)$

In the single-cycle core the multiplier is purely combinational. In the pipeline it is split across two stages — the four partial products are formed in EX (Multiplier stage 1) and combined/shifted in MEM (Multiplier stage 2) — which keeps the multiplier off the critical path.

## 1.2 Signal-Tap instrumentation (IPC & breakpoint debug)

To support on-board IPC measurement and breakpoint debugging the top entity exposes the following registers, all cleared on reset:

- STCNT_o — 8-bit stall counter, increments on every clock in which a stall occurs.
- FHCNT_o — 8-bit flush counter, increments on every clock in which a flush occurs.
- BPADDR_i — 8-bit breakpoint address fed by SW7–SW0; the Signal-Tap trigger fires when IF_PC == BPADDR_i, letting us capture core signals from any PC until the buffer is full.

The instructions-per-cycle figure is verified against the relation

$$IPC = \frac{CLKCNT\_o - (STCNT\_o + 4 + depth \cdot FHCNT\_o)}{CLKCNT\_o} = \frac{InstructionCounter}{CLKCNT\_o}, \quad IPC = 1/CPI$$

# 2. Top-Level Block Review

Both cores are structural top entities. The single-cycle top entity exposes the architectural state and datapath observation ports together with a free-running MCLK cycle counter; the pipelined top entity additionally exposes per-stage PC/instruction taps plus the stall, flush and trigger registers required for IPC and breakpoint debugging. Their I/O is summarised below.

| Port | Single-cycle RV32IM_CORE | Pipelined RV32IM_CORE |
|---|---|---|
| Clock / reset | clk_i (50 MHz → PLL), rst_i (KEY0) | clk_i (50 MHz → PLL), rst_i (KEY0) |
| Debug input | — | BPADDR_i (SW7–SW0) |
| PC / instruction | pc_o, instruction_o | IF/ID/EX/MEM/WB pc_o & instruction_o |
| Control taps | RegWrite_ctrl_o, MemWrite_ctrl_o, Branch_ctrl_o, brTaken_o | (per-stage, internal) |
| Datapath taps | read_data1_o, read_data2_o, write_data_o, alu_res_o | (per-stage, internal) |
| Memory taps | dtcm_addr_o, dtcm_data_wr_o, dtcm_data_rd_o | (internal) |
| Counters | mclk_cnt_o | CLKCNT_o, STCNT_o, FHCNT_o, STRIGGER_o |

**Table 1:** *Top-level port map of the single-cycle and pipelined RV32IM cores.*

Each core instantiates five datapath sub-blocks — IFETCH, IDECODE, CONTROL, EXECUTE and DMEMORY — plus an altpll PLL that generates MCLK and an MCLK counter. The pipelined core adds four pipeline-register banks (IF/ID, ID/EX, EX/MEM, MEM/WB), a combinational Stall-Condition (hazard) unit and a Forwarding unit. The structural decomposition is visible in the RTL Viewer output of Section 3.

## 3. RTL Viewer Results

The Quartus RTL Viewer confirms the structural hierarchy of each core. In the single-cycle netlist the five sub-modules (control:CTL, Ifetch:IFE, Idecode:ID, Execute:EXE, dmemory:MEM), the PLL\G0:MCLK and the mclk_cnt register are clearly separated, with the combinational datapath flowing left-to-right in a single clock. The pipelined netlist shows the same datapath sliced by the four register banks, producing the characteristic columns of edge-triggered registers between stages.

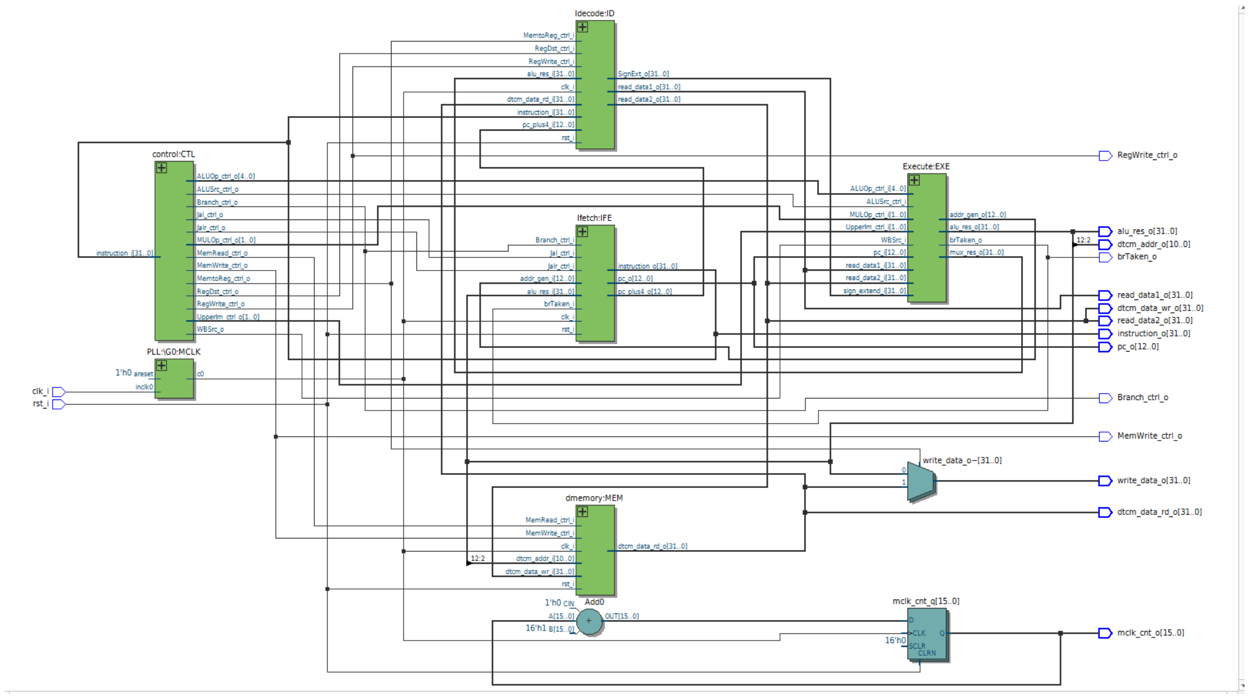



**Figure 1:** *Single-cycle RV32IM — RTL Viewer (structural top level: IFETCH, IDECODE, CONTROL, EXECUTE, DMEMORY, PLL and MCLK counter).*

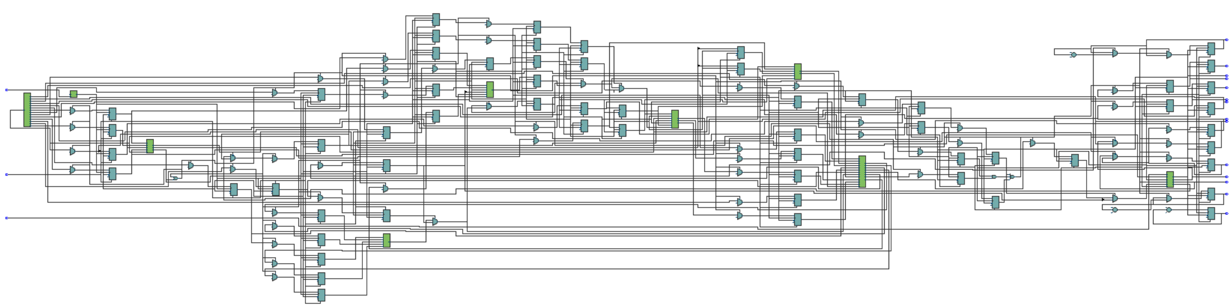


**Figure 2:** *Pipelined RV32IM — RTL Viewer. The added pipeline registers appear as the dense register columns separating the five stages.*

## 4. PPA Characterization

All figures are taken from the same Quartus Prime 25.1std.0 compilation targeting Cyclone V 5CSXFC6D6F31C6 (Timing Models: Final). On Cyclone V the area metric is Adaptive Logic Modules (ALMs) rather than Cyclone-IV logic elements, and the embedded 8-bit multipliers are realised inside variable-precision DSP blocks.

### 4.1 Area

| Resource | Single-cycle | Pipelined |
|---|---|---|
| Logic (ALMs) | 1,448 / 41,910 (3 %) | 1,705 / 41,910 (4 %) |
| Registers (fitter) | 1,314 | 2,024 |
| Registers (synthesis) | 1,279 | 1,864 |
| I/O pins | 270 / 499 (54 %) | 268 / 499 (54 %) |

| Resource | Single-cycle | Pipelined |
|---|---|---|
| Block memory bits | 131,072  (2 %) | 131,072  (2 %) |
| RAM blocks | 14 / 553  (3 %) | 14 / 553  (3 %) |
| DSP blocks (4×8-bit mul) | 4 / 112  (4 %) | 4 / 112  (4 %) |
| PLLs | 1 / 15  (7 %) | 1 / 15  (7 %) |

**Table 2:** *Area characterization (Quartus Fitter). The pipeline costs +18 % ALMs and +54 % registers, driven by the four pipeline-register banks plus the hazard and forwarding logic; memory and DSP usage are unchanged because the ISA and the 16-bit multiplier are identical.*

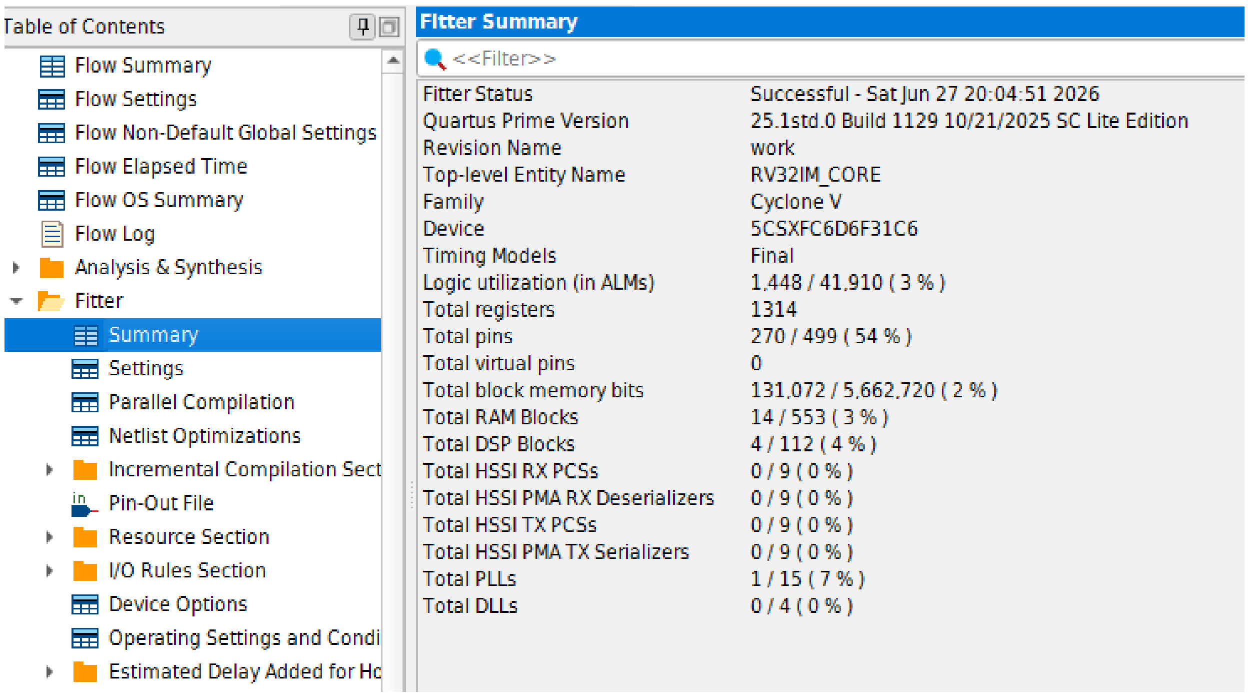



**Figure 3:** *Single-cycle — Fitter resource summary (1,448 ALMs, 1,314 registers, 4 DSP, 1 PLL).*

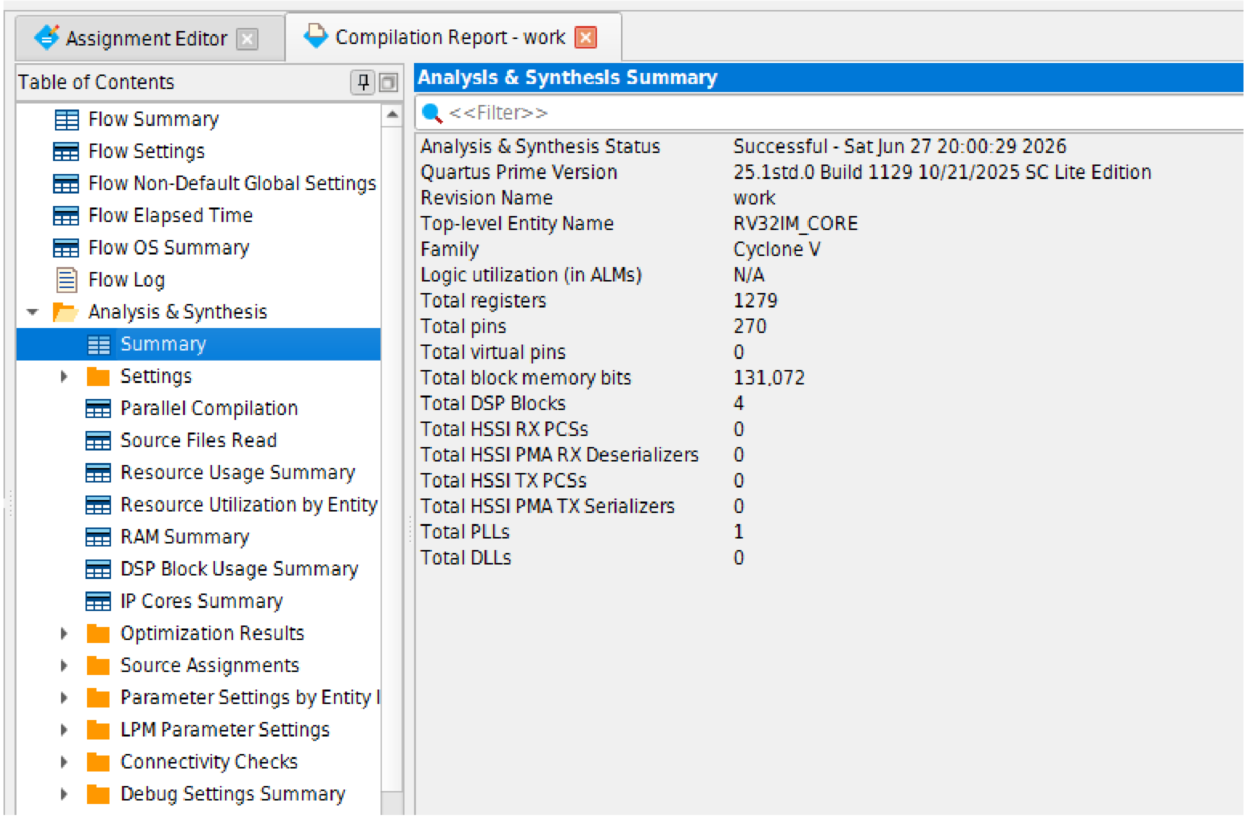



**Figure 4:** *Single-cycle — Analysis & Synthesis summary (1,279 registers, 4 DSP).*

| | |
|---|---|
| Fitter Status | Successful - Sun Jun 28 12:11:10 2026 |
| Quartus Prime Version | 25.1std.0 Build 1129 10/21/2025 SC Lite Edition |
| Revision Name | work |
| Top-level Entity Name | RV32IM_CORE |
| Family | Cyclone V |
| Device | 5CSXFC6D6F31C6 |
| Timing Models | Final |
| Logic utilization (in ALMs) | 1,705 / 41,910 ( 4 % ) |
| Total registers | 2024 |
| Total pins | 268 / 499 ( 54 % ) |
| Total virtual pins | 0 |
| Total block memory bits | 131,072 / 5,662,720 ( 2 % ) |
| Total RAM Blocks | 14 / 553 ( 3 % ) |
| Total DSP Blocks | 4 / 112 ( 4 % ) |
| Total HSSI RX PCSs | 0 / 9 ( 0 % ) |
| Total HSSI PMA RX Deserializers | 0 / 9 ( 0 % ) |
| Total HSSI TX PCSs | 0 / 9 ( 0 % ) |
| Total HSSI PMA TX Serializers | 0 / 9 ( 0 % ) |
| Total PLLs | 1 / 15 ( 7 % ) |
| Total DLLs | 0 / 4 ( 0 % ) |

**Figure 5:** *Pipelined — Fitter resource summary (1,705 ALMs, 2,024 registers, 4 DSP, 1 PLL).*

**Analysis & Synthesis Summary**

<<Filter>>

| | |
|---|---|
| Analysis & Synthesis Status | Successful - Sun Jun 28 12:06:22 2026 |
| Quartus Prime Version | 25.1std.0 Build 1129 10/21/2025 SC Lite Edition |
| Revision Name | work |
| Top-level Entity Name | RV32IM_CORE |
| Family | Cyclone V |
| Logic utilization (in ALMs) | N/A |
| Total registers | 1864 |
| Total pins | 268 |
| Total virtual pins | 0 |
| Total block memory bits | 131,072 |
| Total DSP Blocks | 4 |
| Total HSSI RX PCSs | 0 |
| Total HSSI PMA RX Deserializers | 0 |
| Total HSSI TX PCSs | 0 |
| Total HSSI PMA TX Serializers | 0 |
| Total PLLs | 1 |
| Total DLLs | 0 |

**Figure 6:** *Pipelined — Analysis & Synthesis summary (1,864 registers, 4 DSP).*

## 4.2 Performance

| Metric | Single-cycle | Pipelined |
|---|---|---|
| Fmax (MCLK) | 34.99 MHz | 98.72 MHz (Slow 1100 mV 85 °C) |
| F_MCLK = F_sysclk | ≤ 34.99 MHz (PLL output) | ≤ 98.72 MHz (PLL output) |
| Critical path | Full single-cycle datapath: ITCM → decode/RF read → ALU + result mux chain → DTCM → write-back mux. Slowest sub-module: EXECUTE result-mux chain together with the DTCM RAM access. | ID register-file write/decode path: MEM/WB write-back → RF decoder (Decoder0~22) → RF_q. Slowest sub-module: IDECODE register-file write path. |

**Table 3:** *Performance characterization.*

Pipelining raises Fmax from 34.99 MHz to 98.72 MHz — a 2.82× clock improvement. The theoretical ceiling for an ideally balanced 5-stage pipeline is Fmax_pipe = 5 × Fmax_single = 174.95 MHz; the achieved 98.72 MHz is ≈ 56 % of that ideal. The gap is expected: the stages are not perfectly balanced (the register-file write/decode path and the multiplier stage dominate), and the extra pipeline-register, hazard and forwarding logic adds delay that does not exist in the single-cycle datapath.

| | Fmax | Restricted Fmax | Clock Name | Note |
|---|---|---|---|---|
| 1 | 34.99 MHz | 34.99 MHz | \G0:MCLK\|altpll_component\|auto_generated\|generic_pll1~PLL_OUTPUT_COUNTER\|divclk | |
| 2 | 86.65 MHz | 86.65 MHz | altera_reserved_tck | |

**Figure 7:** *Single-cycle — Fmax: 34.99 MHz on the MCLK (PLL divclk).*

**Slow 1100mV 85C Model Fmax Summary**

<<Filter>>

| | Fmax | Restricted Fmax | Clock Name | Note |
|---|---|---|---|---|
| 1 | 94.12 MHz | 94.12 MHz | altera_reserved_tck | |
| 2 | 98.72 MHz | 98.72 MHz | \G0:MCLK\|altpll_component\|auto_generated\|generic_pll1~PLL_OUTPUT_COUNTER\|divclk | |

**Figure 8:** *Pipelined — Fmax: 98.72 MHz on the MCLK.*

| | Fmax | Restricted Fmax | Clock Name | Note |
|---|---|---|---|---|
| 1 | 90.56 MHz | 90.56 MHz | altera_reserved_tck | |
| 2 | 98.89 MHz | 98.89 MHz | \G0:MCLK\|altpll_component\|auto_generated\|generic_pll1~PLL_OUTPUT_COUNTER\|divclk | |

**Figure 9:** *Pipelined — Fmax at the second slow corner (98.89 MHz on MCLK); 98.72 MHz is taken as the worst case.*

The critical-path technology-map views below locate the slowest cells. For the single-cycle core they sit in the EXECUTE result-mux chain and the data-memory RAM block (interconnect delays ≈ 1.0–1.45 ns per hop). For the pipeline they move into the IDECODE register-file write path (Decoder0~22: IC 1.380 ns; RF_q: IC 2.210 ns).

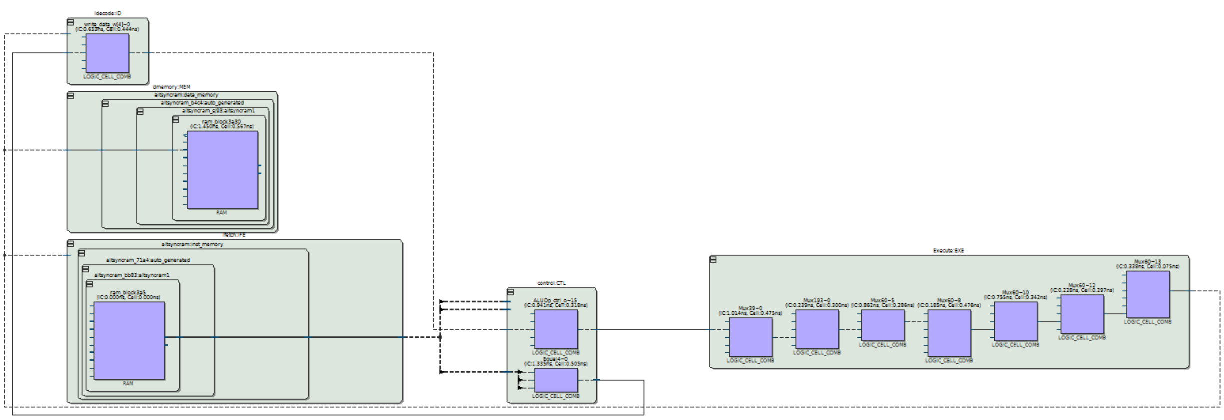


**Figure 10:** *Single-cycle — critical-path technology map: EXECUTE mux chain + data-memory RAM dominate the single-cycle delay.*

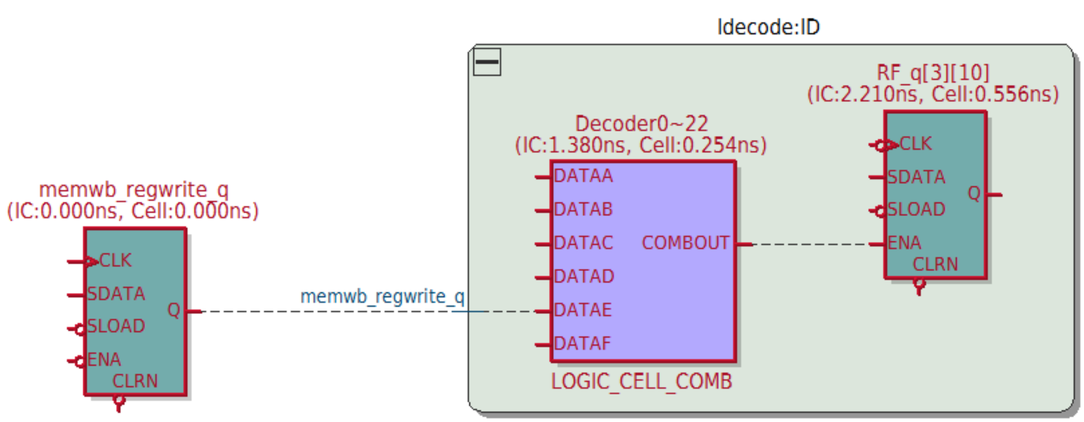


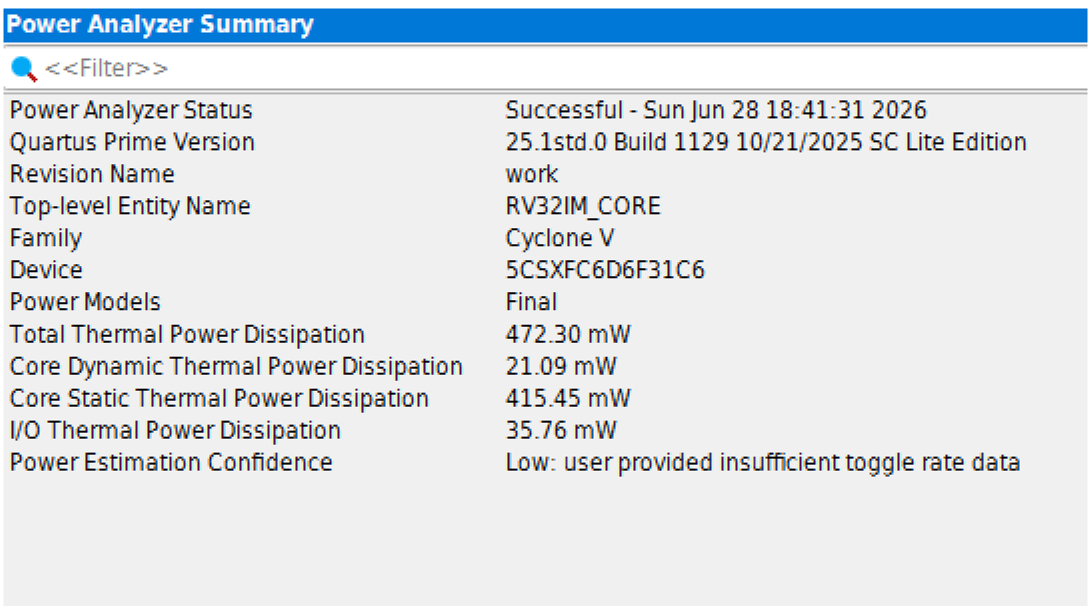

**Figure 11:** *Pipelined — critical-path technology map: the dominant path is the register-file write in IDECODE.*

### 4.3 Power

| Power component | Single-cycle | Pipelined |
|---|---|---|
| Total thermal power | 472.30 mW | 595.04 mW |
| Core static | 415.45 mW | 417.50 mW |
| Core dynamic | 21.09 mW | 88.79 mW |
| I/O thermal | 35.76 mW | 88.76 mW |

**Table 4:** *Power characterization (PowerPlay, Final timing models). Estimation confidence is reported as Low because the .vcd toggle data is partial — the absolute numbers are indicative.*

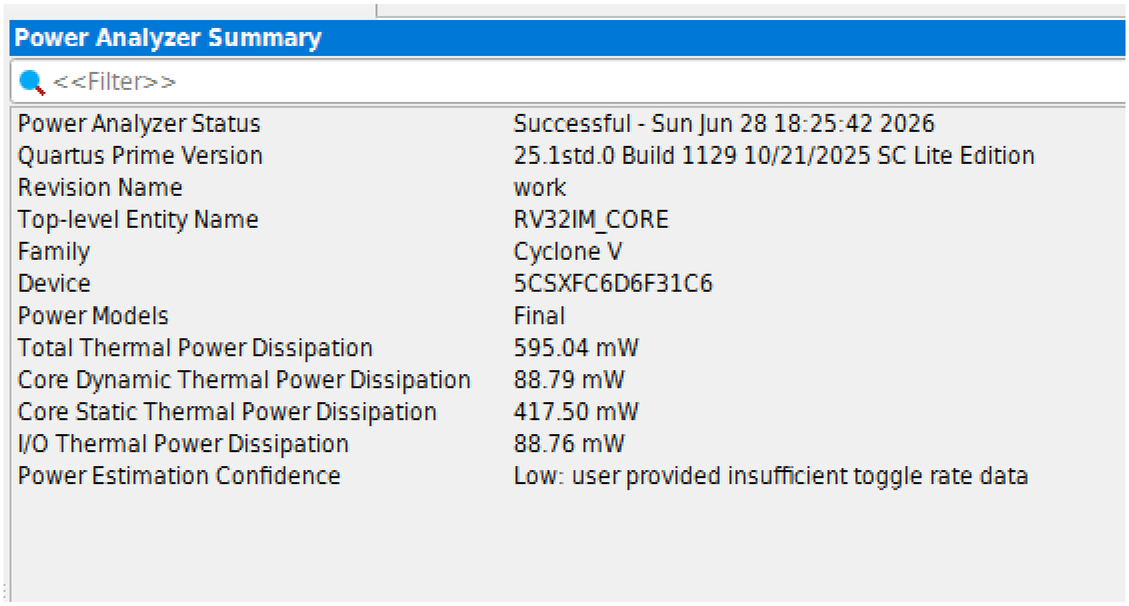
Static power is essentially identical (same device, same process) at ≈ 416 mW. Dynamic power, however, rises from 21 mW to 89 mW: the pipeline holds far more registers (2,024 vs 1,314) and clocks them ≈ 2.8× faster, so switching activity grows accordingly. I/O power also rises (35.8 → 88.8 mW) because the pipeline top entity drives many more toggling observation ports (five per-stage PC and instruction buses plus the counters) for Signal-Tap.


Power Analyzer Summary

<<Filter>>

| | |
|---|---|
| Power Analyzer Status | Successful - Sun Jun 28 18:41:31 2026 |
| Quartus Prime Version | 25.1std.0 Build 1129 10/21/2025 SC Lite Edition |
| Revision Name | work |
| Top-level Entity Name | RV32IM_CORE |
| Family | Cyclone V |
| Device | 5CSXFC6D6F31C6 |
| Power Models | Final |
| Total Thermal Power Dissipation | 472.30 mW |
| Core Dynamic Thermal Power Dissipation | 21.09 mW |
| Core Static Thermal Power Dissipation | 415.45 mW |
| I/O Thermal Power Dissipation | 35.76 mW |
| Power Estimation Confidence | Low: user provided insufficient toggle rate data |

**Figure 12:** *Single-cycle — PowerPlay summary: 472.30 mW total (415.45 static / 21.09 dynamic / 35.76 I/O).*



**Power Analyzer Summary**

| | |
|---|---|
| Power Analyzer Status | Successful - Sun Jun 28 18:25:42 2026 |
| Quartus Prime Version | 25.1std.0 Build 1129 10/21/2025 SC Lite Edition |
| Revision Name | work |
| Top-level Entity Name | RV32IM_CORE |
| Family | Cyclone V |
| Device | 5CSXFC6D6F31C6 |
| Power Models | Final |
| Total Thermal Power Dissipation | 595.04 mW |
| Core Dynamic Thermal Power Dissipation | 88.79 mW |
| Core Static Thermal Power Dissipation | 417.50 mW |
| I/O Thermal Power Dissipation | 88.76 mW |
| Power Estimation Confidence | Low: user provided insufficient toggle rate data |

**Figure 13:** *Pipelined — PowerPlay summary: 595.04 mW total (417.50 static / 88.79 dynamic / 88.76 I/O).*

## 5. Functional Verification and Waveform Analysis

Verification follows the golden-model flow of the assignment: the benchmark application is run in RARS (golden), in ModelSim (RTL simulation) and on the board through Signal-Tap / ISMCE, and the resulting DTCM contents are compared. The benchmark used here writes an incrementing data pattern (1, 2, 3, …) into successive DTCM words, which makes both the data values and the instruction/PC stream easy to follow on the waveforms.

### 5.1 ModelSim RTL simulation

Single-cycle: every instruction completes in one MCLK period, so pc_o advances by 4 each cycle and mclk_cnt_o increments monotonically; alu_res_o and write_data_o produce the stored value in the same cycle the address dtcm_addr_o is presented. Pipelined: the same program is now distributed across the five stages — the per-stage taps (IF/ID/EX/MEM/WB pc & instruction) show one instruction marching down the pipeline each cycle, and STCNT_o / FHCNT_o accumulate the stalls and flushes (here STCNT_o = 08h, FHCNT_o = 07h).

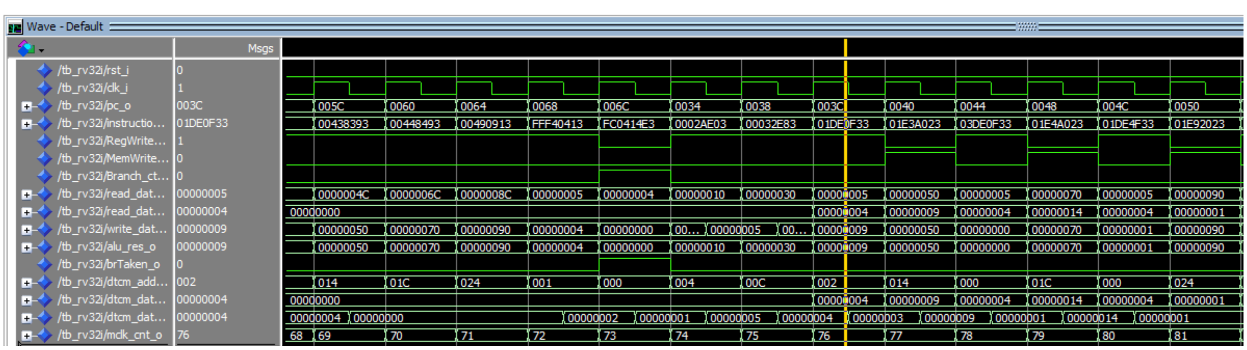



**Figure 14:** *Single-cycle — ModelSim waveform of the benchmark (one instruction per MCLK; PC +4 each cycle).*

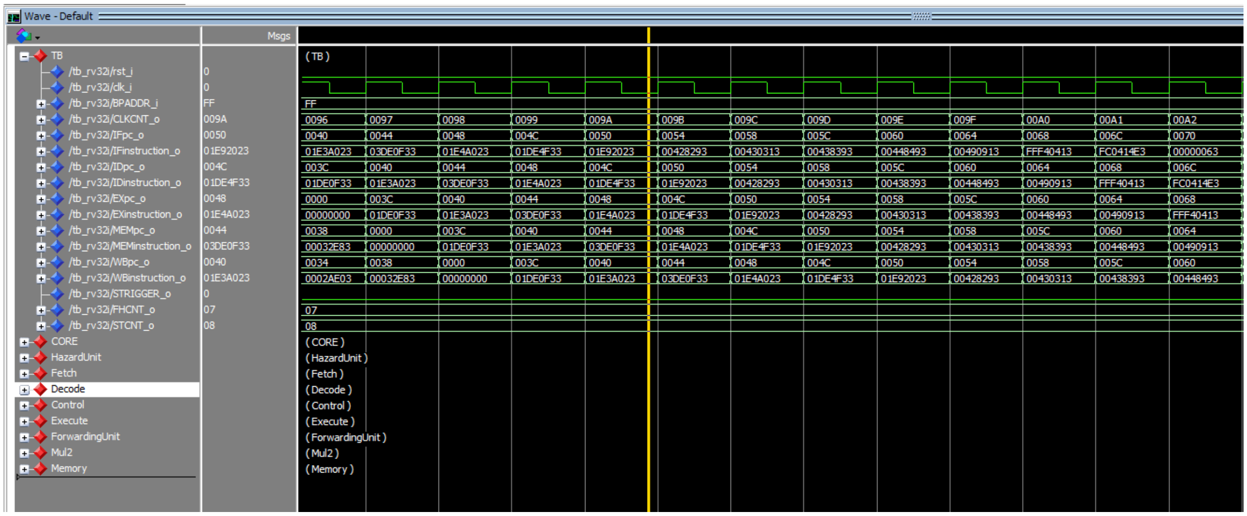



**Figure 15:** *Pipelined — ModelSim waveform of the benchmark, showing the per-stage PC/instruction taps and the STCNT/FHCNT counters.*

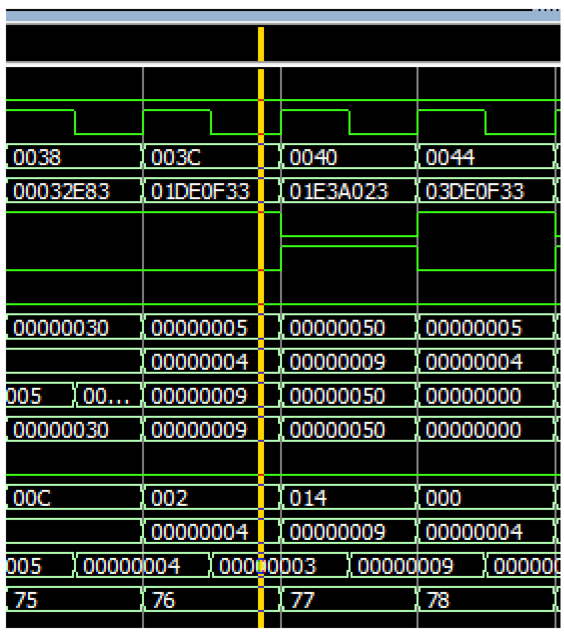



**Figure 16:** *Single-cycle — add instruction.*

## 5.2 On-board Signal-Tap validation

On the DE-SoC board the Signal-Tap trigger is armed at IF_PC == BPADDR_i and the core is single-stepped with Auto-run. The single-cycle captures below show the start of the program (PC sweeping 0x00→) and the closing loop branch, where Branch_ctrl_o and brTaken_o assert. The pipelined capture shows all per-stage PC/instruction taps advancing together with CLKCNT_o, STCNT_o = 08h and FHCNT_o = 07h, exactly matching the ModelSim run.

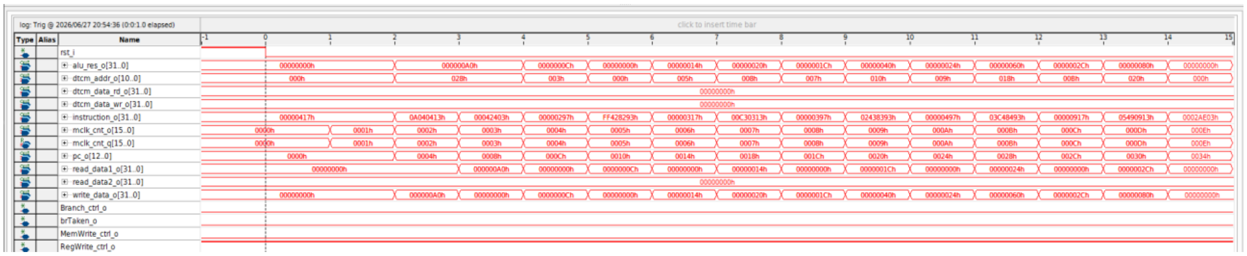


**Figure 17:** *Single-cycle — Signal-Tap capture at program start (PC sweep, one instruction per cycle).*

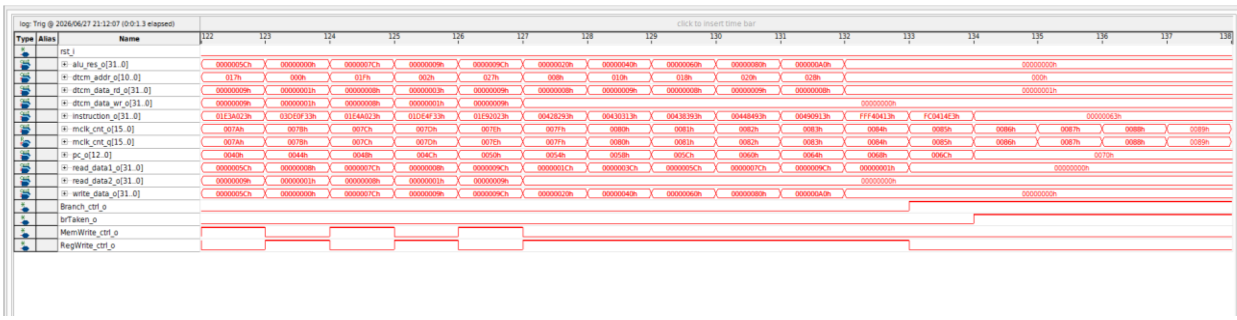


**Figure 18:** *Single-cycle — Signal-Tap capture near program end: Branch_ctrl_o / brTaken_o assert on the loop branch.*

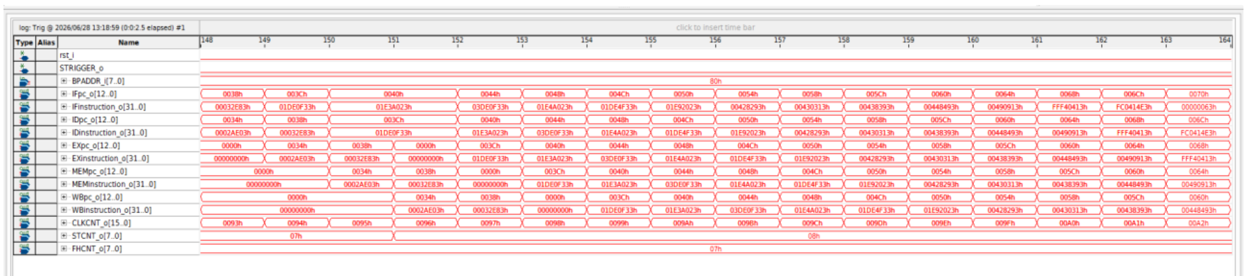


**Figure 19:** *Pipelined — Signal-Tap capture: per-stage PC/instruction taps, CLKCNT_o, STCNT_o = 08h, FHCNT_o = 07h, BPADDR_i = 80h.*

**Figure 20:** *Single-cycle — Signal-Tap instance manager (auto_signaltap_0 on DE-SoC, capturing the core observation ports).*

## 6.3 Golden-model memory comparison (ISMCE)

After the run the ITCM and DTCM contents are read back with the In-System Memory Content Editor and compared against the RARS DTCM.hex / DTCM.mem golden files. The ITCM holds the assembled program image; the DTCM holds the incrementing result pattern. Both cores produce byte-identical DTCM contents, which closes the golden-model check.

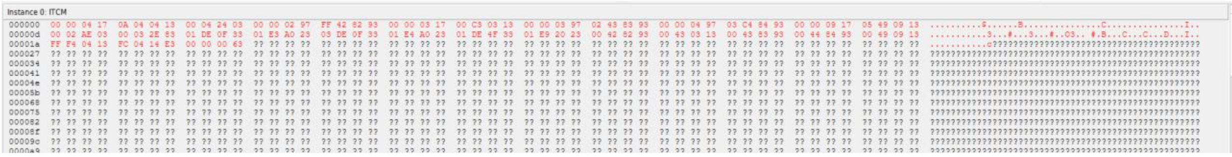


**Figure 21:** *Single-cycle — ITCM contents read back via ISMCE (assembled program image).*

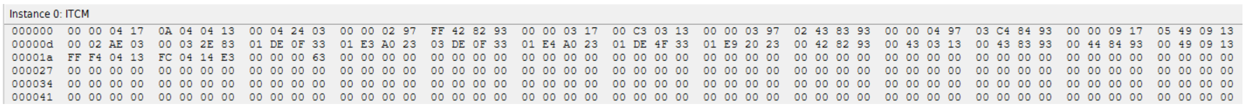


**Figure 22:** *Pipelined — ITCM contents (identical program image).*

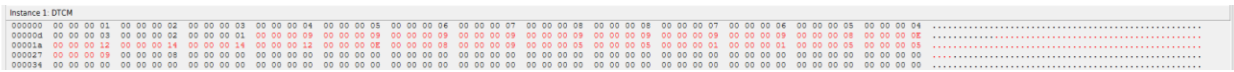


**Figure 23:** *Single-cycle — DTCM contents: the incrementing result pattern written by the benchmark.*

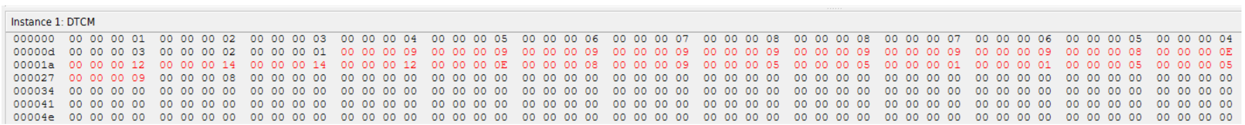


**Figure 24:** *Pipelined — DTCM contents: byte-identical to the single-cycle and to the RARS golden model.*

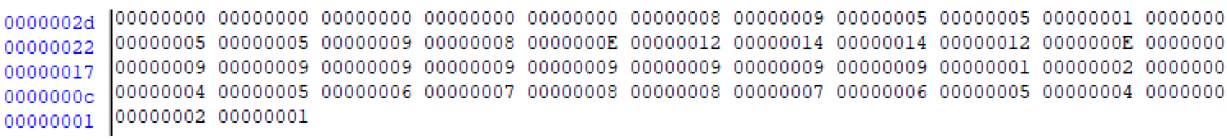


**Figure 25:** *Pipelined — DTCM.mem golden dump (ModelSim) used as the reference for the comparison.*

*Per the assignment, basic waveforms for benchmark applications test1–test4 are required. The captures above correspond to the representative benchmark; the remaining test1–test4 waveforms should be inserted here in the same SC-over-pipeline format once captured.*

### 5.4 IPC analysis

For the single-cycle core there are no stalls or flushes (STCNT = FHCNT = 0), so CPI = 1 and IPC = 1 by construction. For the pipeline, IPC is recovered from the on-chip counters using the relation of Section 1.2. With the captured values (STCNT_o = 8, FHCNT_o = 7, flush depth = 3) the executed-instruction count is CLKCNT_o − (8 + 4 + 3·7) = CLKCNT_o − 33; at the capture point CLKCNT_o ≈ 162 this gives ≈ 129 instructions and IPC ≈ 0.80 (CPI ≈ 1.25). The two sides of the IPC equation match the ModelSim instruction count, confirming correct stall/flush accounting. The final IPC must be read from the end-of-program CLKCNT_o.

## 6. Conclusions

Both RV32IM cores were verified functionally identical (matching DTCM golden model in RARS, ModelSim and on-board ISMCE) while exhibiting the expected PPA trade-off. Pipelining raised Fmax 2.82× (34.99 → 98.72 MHz) at a cost of +18 % ALMs, +54 % registers and ~4× dynamic power, with the critical path migrating from the single-cycle EXECUTE/DTCM combinational path to the pipelined register-file write path. The achieved Fmax reaches ~56 % of the ideal 5× ceiling, limited by stage imbalance and the added hazard/forwarding/register overhead — a textbook illustration of the single-cycle versus pipelined design space.